



# DEV PATRA

CHEMICAL & AI/ML RESEARCHER

## PROFILE

I am a Chemical Engineering student and a GATE 2026 qualified candidate with strong skills in AI/ML, research, and data-driven problem solving. I also have hands-on experience in Life Cycle Assessment (LCA). I am a quick learner who values teamwork and focuses on delivering meaningful results.

I have worked on developing machine learning-based biochar models at DRDO, optimizing industrial control systems at RPDS, and collaborating during my internship at Dow Chemical International Pvt. Ltd., where I was awarded 1st Runner-Up. I aim to build practical solutions and contribute effectively to every project I take on.

## WORK EXPERIENCE

### DOW CHEMICAL COMPANY

RESEARCH & DEVELOPMENT INTERN

01/06/2025 - 31/08/2025

- Conducted a systematic review and analysis of 500+ patents and research publications to identify the most suitable LCA tools.
- Mapped relevant databases for different systems to support sustainability assessment of chemical processes.

### RELIABLE PROCESS DESIGN SOLUTION

DATA ANALYTICS INTERN

01/03/2024 – 30/06/2025

- Developed intuitive GUIs using Tkinter and web apps with Streamlit, enhancing user experience for industrial tasks.
- Worked on implementation of NN-MPC, traditional MPC, and optimized PID control systems in batch polymerization reactors, resulting in a faster cycle time.
- Achieved 70% improvement in setpoint settling time through optimization of PID using Differential Evolution algorithms.
- Conducted comprehensive fault analysis on the Tennessee Eastman Process using AI/ML, identifying around 20 faults with high accuracy.

### DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(DIAT), DRDO

RESEARCH INTERN

01/09/2023 – 31/10/2023

- Developed ML and AI-based framework to predict the removal efficiency of heavy metals from industrial wastewater using biochar.
- Designed and tested 12 hybrid ANN-metaheuristic models and 22 ML models, including SVM, GPR, and ensemble methods like LSBoost.
- Implemented 10+ nature-inspired optimization algorithms to fine-tune ANN predictions and compare modelling strategies.

## CONTACT

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Jalgaon, Maharashtra

<https://devpatra07.github.io/>  
(Personal Visit for Recent updates).

## EDUCATION

2021 - 2026

INSTITUTE OF CHEMICAL TECHNOLOGY,  
MUMBAI

- (B.tech + M.tech) Integrated Degree in Chemical Engineering
- GPA: 9.9 / 10

## SKILLS

- Research & Technical Project Execution
- Scientific Communication
- Collaborative Teamwork
- Deadline-Oriented Execution
- Technical Leadership
- Scientific Writing
- Data-Driven Problem Solving

## LANGUAGES

- English (Fluent)
- Hindi (Fluent)
- Marathi (Intermediate)
- Bengali (Native)

## EDUCATION

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### Institute of Chemical Technology, Mumbai

- (B.tech + M.tech) Integrated Degree in Chemical Engineering
- GPA: *9.9 / 10*     [View Certificate](#)

### Chhatrapati Shivaji Junior College

- HSC
- Percentage: *89.83%*

### St. Teresa's Convent Higher Secondary School, Jalgaon, Maharashtra.

- SSC
- Percentage: *86.80%*

## EXTRACURRICULAR

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- **Web Design Head & Treasurer** – Techfest AAKRITI 1.0: Led web design and managed finances, ensuring smooth execution of the tech festival.
- **Student Editor** – MARJAL (ICT Mumbai, MARJ Campus): Managed content and editorial team for the biannual magazine, ensuring high-quality publication.

## PROJECTS

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NAMoSTe Software (Novel Atomic & Molecular Structure Explorer)

(Intellectual Property Secured; Registered with the Copyright Office, Government of India – Extracts from the Official Register of Copyrights) NAMoSTe is a GUI-based software for generating repeating polymer structures in 1D, 2D, and 3D

Link: [View Certificate](#)

## CONFERENCES AND SEMINARS

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International Conference on Machine Learning and Data Engineering.

Link: [View Certificate](#)

## HONOURS AND AWARDS

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1. Dr. Tobias Rosenthal Best Student Award (2026) – Baerlocher India
2. 2<sup>nd</sup> Best Project Internship in Dow Inc. for Project titled Life Cycle in Chemical Industry.
3. AIR 950 (First Attempt)- GATE 2026 (Chemical Engineering)
4. State-level athlete in both Table Tennis and Roll Ball.